

NATURAL GAS DELIVERY OPTIMIZATION

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ML forecasting + stochastic linear programming to minimize cash-out penalties across daily injection/withdrawal and end-of-month storage constraints.

DATASET

3,410 days

PERIOD

Aug 2015 - Nov 2024

KEY RESULTS

42.3%

Penalty Cost Reduction

ACTUAL PENALTY

\$532,491

420 violation days

OPTIMISED PENALTY

\$306,978

156 violation days

PENALTY SAVED

\$225,513

across 3 years

VIOLATION REDUCTION

62.9%

264 fewer days

TOTAL COST ACTUAL

\$11,863,466

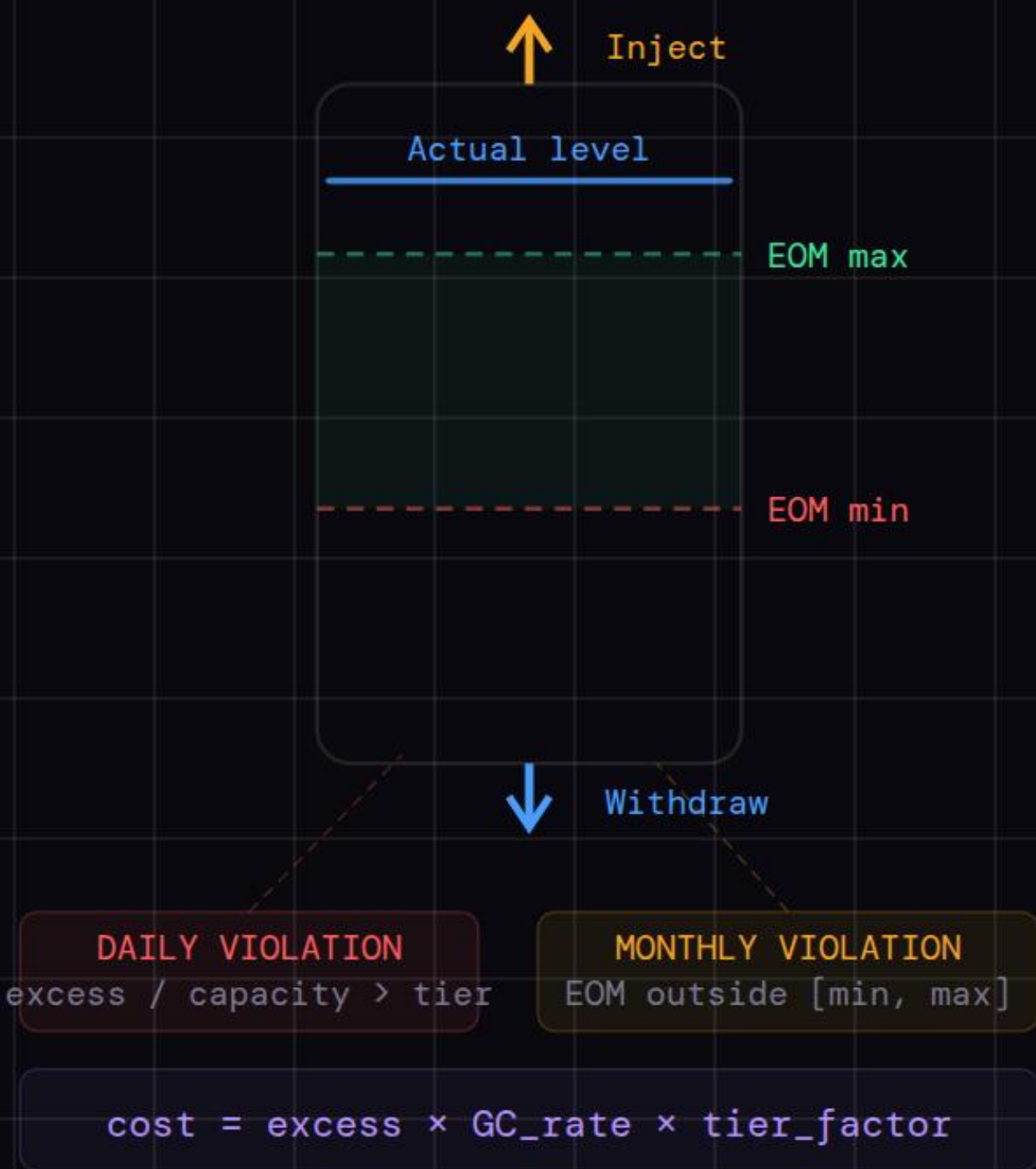
base + penalty

TOTAL COST OPT

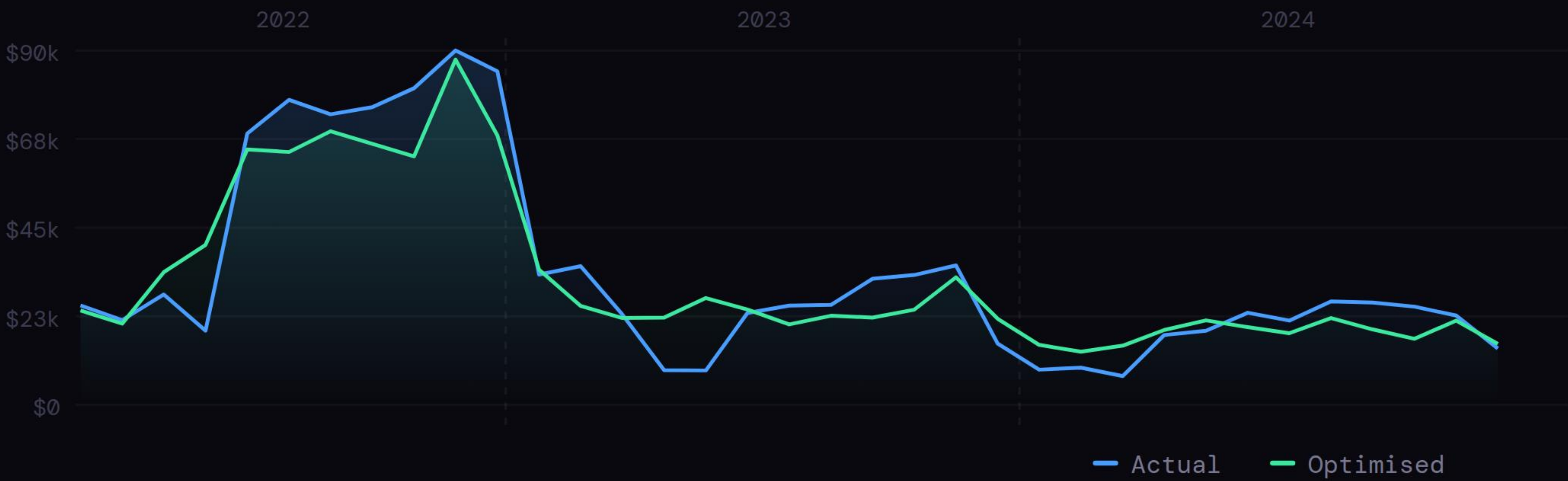
\$11,396,756

\$46,671 saved (3.9%)

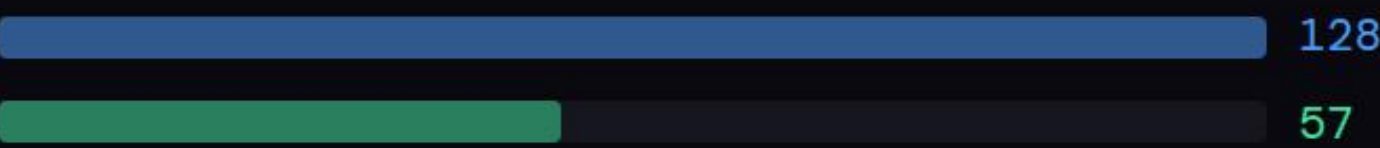
PENALTY EVALUATION



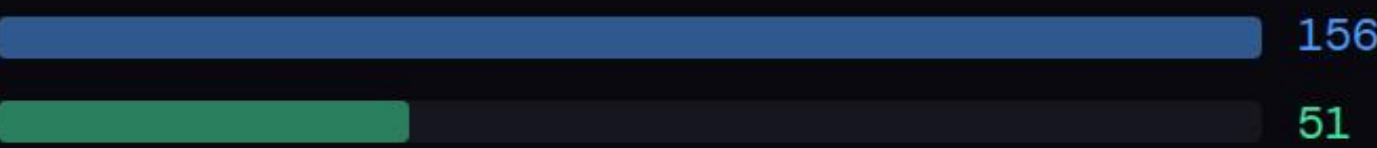
MONTHLY COST - ACTUAL VS OPTIMISED



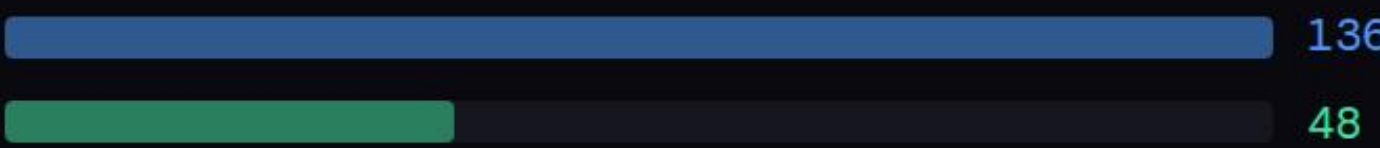
2022 - Violations



2023 - Violations



2024 - Violations



METHODOLOGY

PIPELINE

01 2-Day-Ahead Blended Forecasting

Blended quantile model (70% GBM + 30% QuantileRegressor) for P10/P50/P95. Features: lags T-3 to T-30, rolling stats, holiday flags, storage %, days-to-EOM. Minimum lag = 2 enforces no leakage.

GBM+QR

02 Quantile Uncertainty Bands

Three quantile levels (P10, P50, P95) trained simultaneously. P95 forecast used as conservative usage estimate for LP. Monthly expanding-window retrain across 36 model snapshots (Jan 2022 - Dec 2024).

P10/P95

03 Adaptive Contract Limit Buffering

Month-specific injection buffers (78-85%) and withdrawal buffers (80-88%) shrink effective LP limits. Tighter in summer (inj) and April (wd) where violations historically concentrate.

Adaptive

04 Cost-Weighted Deterministic LP

Minimises $\sum c[t] \cdot \text{delivery}[t]$ where $c[t] = 1 - \alpha \cdot (\text{inj_norm} + \text{wd_norm})$, $\alpha=0.10$. Hard flow/storage bounds, soft multi-EOM and mid-month steering constraints. HiGHS solver, 3-level fallback.

HiGHS

05 Rolling MPC Loop (60-day horizon)

60-day lookahead, act on delivery[0] only. Overshoot guard floors delivery when storage exceeds EOM_max. Hard post-LP clip to raw contract limits. Re-plan every day with updated actuals.

MPC

ROLLING LP STRUCTURE (V4)

Forecast Input

usage_p95[t] horizon=60d

LP Solver (HiGHS)

$\min \sum c[t] \cdot \text{delivery}[t]$
 $c[t] = 1 - \alpha \cdot (\text{inj_norm} + \text{wd_norm})$

var: $\text{delivery}[t] \geq 0$, $t=1..H$
solver: HiGHS | $\alpha = 0.10$

Act on delivery[0] only

re-plan daily · MPC loop

HARD CONSTRAINTS

$\text{delivery}[t] \leq \text{inj_lim}[t] \cdot \beta_{\text{inj}} + \text{usage}[t]$
 $\beta_{\text{inj}} \in [0.78, 0.85]$ by month

$\text{delivery}[t] \geq \text{usage}[t] - \text{wd_lim}[t] \cdot \beta_{\text{wd}}$
 $\beta_{\text{wd}} \in [0.80, 0.88]$ by month

$0 \leq \text{storage}[t] \leq \text{Max Storage}$
 $\text{storage}[t] = s_0 + \sum \text{delivery} - \sum \text{usage}$

SOFT CONSTRAINTS

$\text{EOM_min} \leq \text{storage}[\text{eom}] \leq \text{EOM_max}$
one per month in horizon · buf=6-10%

$\text{storage}[\text{day15}] \approx (\text{EOM_min} + \text{EOM_max}) / 2$
slack = ±6%·capacity mid-month steer

POST-LP ADJUSTMENTS

Overshoot Guard
floor delivery if
storage > EOM_max

Hard Clip
clip to raw contract
inj / wd limits

HORIZON

60d

LAG

2-day

QUANTILE

P95

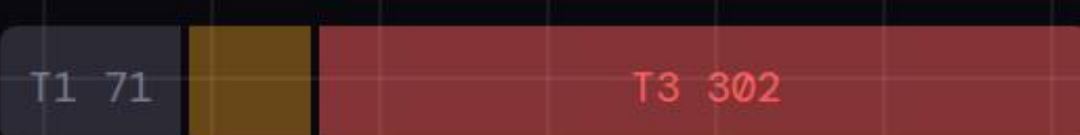
LP MARGIN A

0.10

CASH-OUT TIERS

TIER	VARIANCE	PENALTY
T1	0-10%	0%
T2	10-20%	15%
T3	>20%	40%

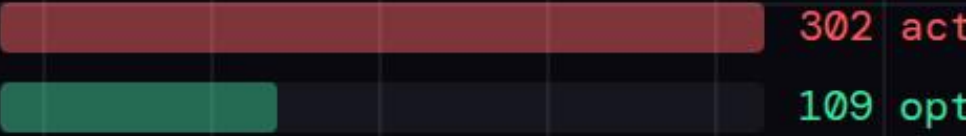
TIER DISTRIBUTION - ACTUAL



TIER DISTRIBUTION - OPTIMISED



TIER 3 DAYS



FORECAST ACCURACY

89.0%

[P10, P95] interval coverage

P50 FORECAST MAE - 2-DAY AHEAD

229.6 units

vs mean daily usage of 1,736.7 — 13.2% relative error

P10-P95 INTERVAL WIDTH

±515 units

P95 over-forecasts by +538 on average — conservative by design

MODEL DETAILS

Target	Total Usage (direct)
Min lag	T-3 (lag_1 + 2-day shift)
Eff. lags	T-3,4,5,9,16,23,30
Features	23 total incl. holiday flags
Retrain	36 snapshots, monthly expanding